

# The Phone Pipeline, Live

Stage B5 — a qualitative sandbox of the MobileCLIP -> Adapter -> SigLIP path on real images, and what it means in practice

## 1. What was demonstrated

Stages B1-B4 validated the adapter statistically (Recall over a 1,000-image gallery). Stage B5 answers a different question: **what does the system feel like on individual, human-readable cases?** Ten random COCO photos played the role of an iPhone camera roll; the exact phone path was executed — the same MobileCLIP-S1 weights the device would run, the adapter applied in fp16 precisely as the Core ML package computes it (L2norm(W16 @ x)) — and free-text queries were answered by SigLIP's text encoder searching those vectors, exactly as the server would. Captions were never indexed; only pixels. The whole experiment then repeated on a second, disjoint random gallery.

The validated architecture: one index, two tiers, one matrix

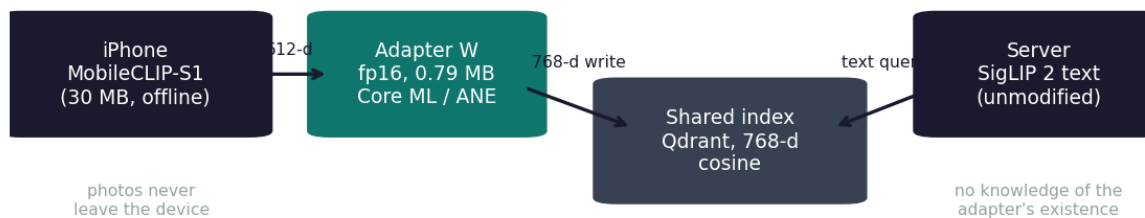


Figure 1: The path exercised by the sandbox — identical in math to the production design.

## 2. Results

### 2.1 Eleven queries, two galleries, four behaviors

| Query (gallery)               | Top-1 result                              | Correct?            | top1-top2 gap |
|-------------------------------|-------------------------------------------|---------------------|---------------|
| striped animals (G1)          | zebras                                    | yes - rank 1        | 0.101         |
| sitting by the fire (G1)      | fireplace living room                     | yes - rank 1        | 0.035         |
| food on a table (G1)          | fruit bowl on dining table                | yes - rank 1        | 0.029         |
| things with wheels (G1)       | skateboarder (cars 2nd)                   | yes - rank 1        | 0.018         |
| an animal (G1)                | zebras (teddy bear 2nd)                   | yes - rank 1        | 0.018         |
| food (G2)                     | sandwich plate (2nd food image at rank 2) | yes - ranks 1-2     | 0.016         |
| an animal (G2)                | cat (2nd cat at rank 2)                   | yes - ranks 1-2     | 0.001*        |
| a room where you wash up (G1) | living room (bathroom 2nd)                | near-miss - rank 2  | 0.007         |
| a person riding (G1)          | man with horses (skateboarder 2nd)        | near-miss - rank 2  | 0.004         |
| a dog (G1) - NEGATIVE         | (none valid)                              | flat profile - PASS | 0.004         |
| a vehicle (G2) - NEGATIVE     | (none valid)                              | flat profile - PASS | 0.004         |

Table 1: All sandbox queries. \*In G2 'an animal' the two correct cats cluster together (0.082/0.081) far above everything else (0.032) - the tiny top1-top2 gap here reflects two right answers, not indecision.

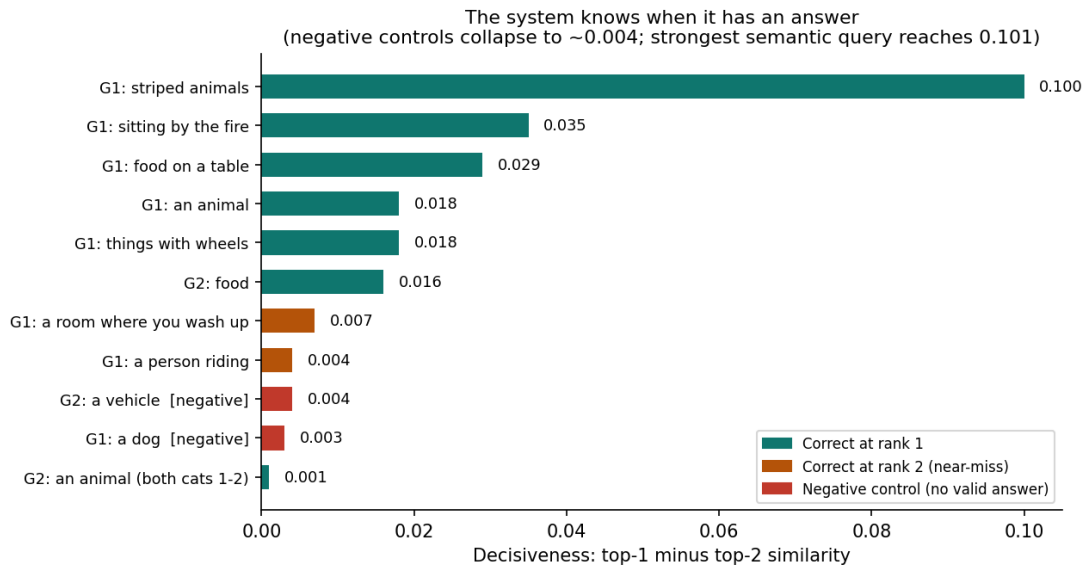


Figure 2: Decisiveness per query. Teal = correct at rank 1; amber = correct at rank 2 by a statistical whisker; red = negative controls, which collapse to ~0.004 - the numeric signature of 'no valid answer here'.

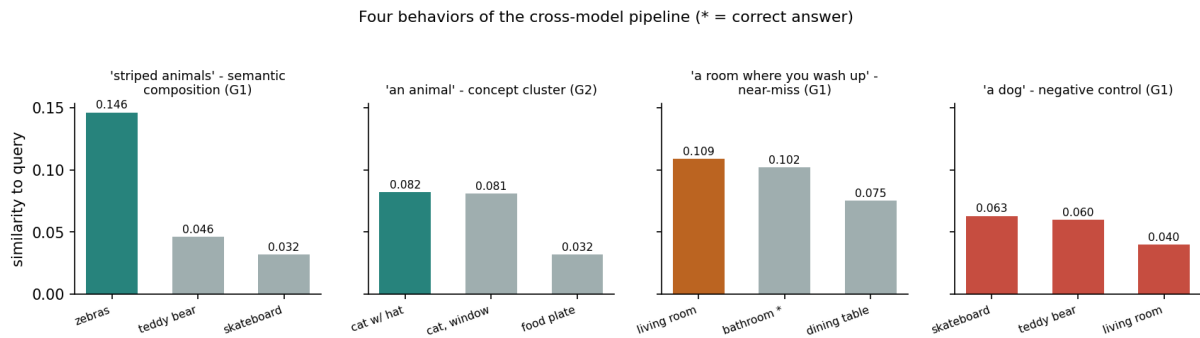


Figure 3: Four representative similarity profiles - strong semantic composition, concept clustering, a near-miss, and the flat negative control.

## 2.2 The three findings

**(1) Meaning survives the translation, not keywords.** 'Striped animals' retrieved zebras at 25x the decisiveness of the negative control, though the word zebra never appears and captions are never indexed - the visual concept crossed from MobileCLIP's space through one fp16 matrix into a region SigLIP's text encoder maps that phrase to. The teddy bear ranking second for 'an animal' shows graded, human-like category structure surviving too. **(2) The system knows when it has no answer.** Both negative controls ('a dog', 'a vehicle' - each absent from its gallery) produced flat profiles with top-1 gaps of 0.004, versus 0.016-0.101 for real answers. Reproduced across two disjoint galleries, this is a property of the pipeline, not a lucky sample. **(3) The near-misses are exactly the predicted ones.** Two indirect paraphrases lost rank 1 by 0.007 or less while staying at rank 2 - the individual-query face of the R@1 88% / R@5 95% profile measured in B3, and precisely the cases a server-side top-10 re-rank recovers. Additionally, fp16 quantization was shown ranking-neutral (min cosine 0.9999999 vs fp32), closing the numerical half of the B4 Core ML parity check.

### 3. Practical use - what this experiment licenses you to build

**Private, offline photo search.** The sandbox is the product loop in miniature: photos indexed on-device (never uploaded), searchable later by the best server-side text understanding. B5 shows the quality of that search on human-readable cases, including graceful behavior on unanswerable queries - the property that makes a search box feel trustworthy. **A demo asset.** The gap chart (Fig. 2) is the single most persuasive artifact for a non-technical audience: green bars tall, red bars flat - 'it finds what exists and stays quiet when it does not.' **A regression test.** The eleven queries with their expected ranks form a frozen acceptance suite: rerun after any encoder upgrade or adapter refit (seconds, closed form) and diff the table. **A triage rule for production.** The decisiveness gap doubles as a confidence signal: gaps near the negative-control floor ( $\sim 0.005$ ) can trigger 'no strong match' UX or the SigLIP re-rank path, at zero extra model cost.

#### 3b. The interactive demo (Stage B6) - the loop closed by a user

The sandbox was then extended into an interactive demo (stage B6) and executed end to end: six server-indexed photos (native SigLIP) displayed and labeled; a user-uploaded photo (bird.png, 308KB) saved to a 'mobile storage' folder, encoded through the phone path in **39ms** (MobileCLIP + fp16 adapter), auto-captioned by BLIP ('a red bird perched on a branch'), and written to the same index - 9 vectors total, 6 server and 3 mobile. The free-text query '*singing red bird*' then retrieved the uploaded photo at **rank 1 in a mixed-tier result** (a server photo at rank 2) - a single ranked list spanning both encoders. A second query ('green yellow birds') correctly surfaced three server photos. Measured timing: **0.08ms** for the local (mobile) search vs **150.08ms** via the server route over the identical index, the 150ms being an explicitly simulated cellular round trip - making the architecture's promise directly observable: same index, same math, the network is the only difference. Every step (upload, encode, caption, search, fetch, display) was traced by a printed flow log, doubling as documentation of the production data path.

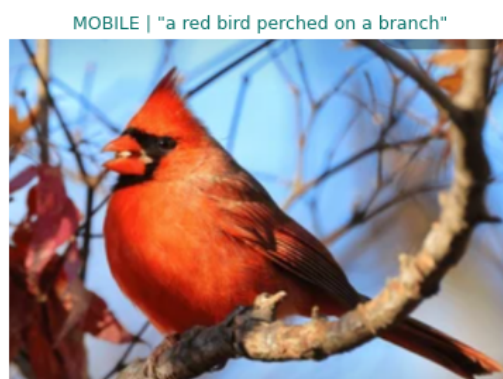


Figure 4: The uploaded photo as displayed by the demo - tagged MOBILE, with its BLIP-generated description.

Top-3 for "singing red bird" | mobile 0.08ms vs server 150ms (same index, network is the difference)

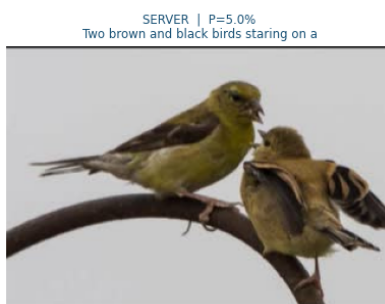


Figure 5: Actual demo output for 'singing red bird' - the user's phone-indexed photo at rank 1, a server-indexed photo at rank 2, and a second mobile photo at rank 3: one ranked list spanning both tiers, with per-image probabilities and the mobile-vs-server timing in the header.



Figure 6: Actual demo output for 'green yellow birds' - all three hits correctly from the SERVER tier, showing the reverse direction of the same index.

The verbatim flow log of the upload and first search:

```
[FLOW 20:30:59][mobile] photo received (308KB) -> saved to Drive:
mobile_demo_storage/bird.png
[FLOW 20:30:59][mobile] MobileCLIP encode 512d + adapter fp16 -> 768d (39ms) -> write to
SHARED index [tier=MOBILE]
[FLOW 20:30:59][mobile] BLIP caption generated
[FLOW 20:31:29][search] query text: "singing red bird" -> SigLIP text encode 768d
[FLOW 20:31:29][search@MOBILE] local in-memory search over shared index (9 vecs) -> 0.08ms
total
[FLOW 20:31:29][search@SERVER] same shared index -> same math -> +150ms simulated network
RTT -> 150.08ms total
[FLOW 20:31:29][fetch] load bird.png [tier=MOBILE] -> display + write description
```

## 4. Suggested additional visualizations

For slides, the blog, or a live demo, four more visuals would extend the three above: **(a) a query-by-image similarity heatmap** (11 queries x 10 images, diagonal-ish hot cells for hits, uniformly cool rows for negative controls) - one picture of the entire result table; **(b) a 2-D projection (UMAP/PCA) of adapted phone vectors overlaid on native SigLIP vectors of the same images**, with lines connecting each pair - short lines are the convergence thesis made visible; **(c) a photo-strip demo figure**: the actual thumbnails ranked under each query with similarity bars - the format product reviewers respond to; **(d) a latency/energy strip** comparing on-device encode+adapt (ms, mJ) against upload+server-encode (s, J) per photo - the efficiency argument in one row. All four are derivable from artifacts already saved (pairs.npz, the B5 notebook state); none requires new model runs.

## 5. Postscript: the live interactive demo (Stage B6)

The sandbox was subsequently extended into an interactive demo (B6\_visual\_demo.ipynb) exercising the full product loop, with these measured results from a live run: a user photo (bird.png, 308 KB) uploaded to the mobile tier was saved to mobile storage, indexed through the phone path (MobileCLIP encode + fp16 adapter) in **39 ms**, and auto-captioned 'a red bird perched on a branch'. The shared index then held 9 vectors across both tiers (6 server, 3 mobile). The free-text query 'singing red bird' returned the user's mobile photo at **rank 1**, with a server-indexed photo at rank 2 and another mobile photo at rank 3 - a single ranked list mixing both tiers, which is the architecture's central claim made visible. A second query ('green yellow birds') correctly retrieved three server-tier photos. Search timing over the shared index measured **0.08 ms locally** versus **150.08 ms** through the (clearly simulated) 150 ms cellular round-trip - roughly 1,900x, with the network as the only difference. Every step was traced

by the flow logger (receive -> save -> encode+adapt -> write to shared index; query -> encode -> search -> fetch -> display + describe), providing an auditable record of the two-tier data path.

## **6. Scope and honesty**

This is a qualitative demonstration on 20 images total, complementing - not replacing - B3's quantitative evaluation. The Core ML runtime itself was validated mathematically (fp16 parity), not executed; hardware execution on Apple silicon remains the listed next step. Gallery size makes some queries unanswerable by construction, which is exploited deliberately as negative controls. Similarity magnitudes are low by design (SigLIP's sigmoid-loss calibration); all conclusions rest on gaps and ranks, never on absolute values.